

PITWALL

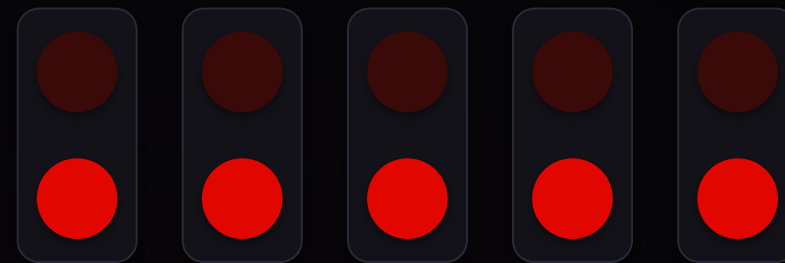
Predicting lap times, tyre wear and the pit call from 125,046 laps of Formula 1.

LAP TIME

TYRE WEAR

BOX OR STAY OUT

WHAT IF



ARNAV YADAV • 2026-10-06

EVERY LAP, *ONE QUESTION.*

“Box, or stay out?”

A pit stop costs about 22.6 seconds. Old tyres cost a little more every lap. The pit wall has to find the lap where fresh tyres win back more than the stop costs, knowing a Safety Car could make the stop almost free.

01

PREDICT

the lap time

02

MODEL

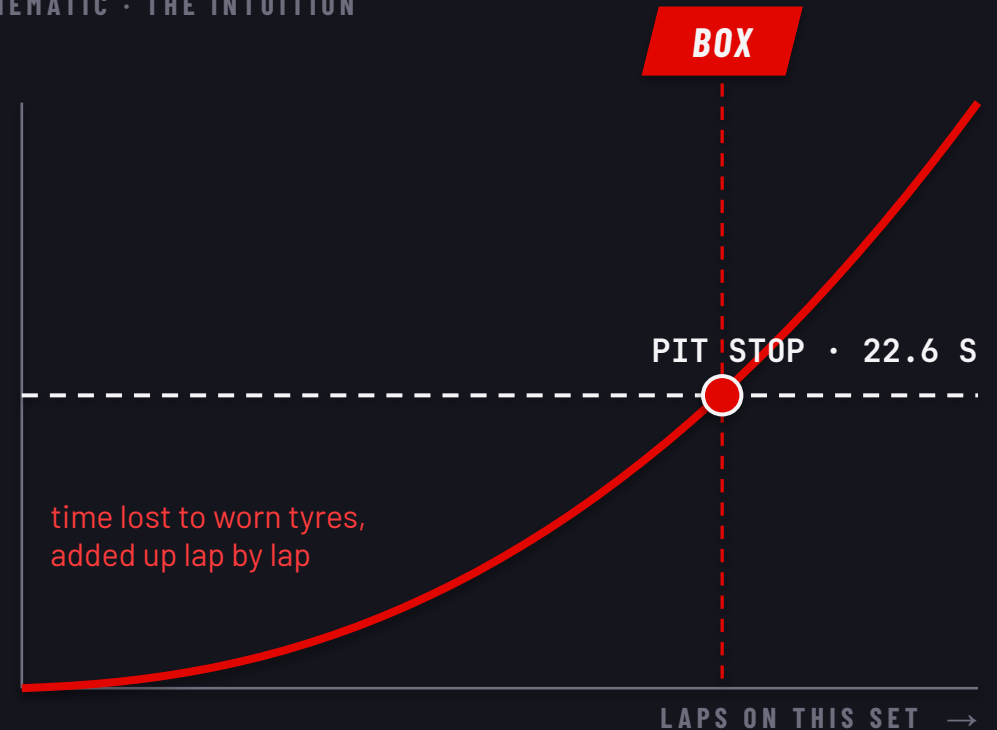
the tyre

03

DECIDE

the call

SCHEMATIC · THE INTUITION



The engine solves the whole remaining race, Safety Car odds included.

— THE DATA

FIVE SEASONS. *ONE HONEST TEST.*

114

RACES

125,046

LAPS

26

CIRCUITS

5

SEASONS · 2021-2025

FIT · 2021-2023

Model parameters learned only from these three seasons.

TUNE · 2024

Every knob, threshold and prior chosen here.

TEST · 2025

Touched once, after everything was frozen.

Every lap kept: in-laps, out-laps and Safety Car laps are exactly what a pit-stop model needs. Representative laps (90,709) are flagged for fitting pace.

IT LOOKED **GREAT.**

RIDGE REGRESSION ON LAP TIME · 71 FEATURES

R^2 **0.91**

on the 2024 dev season. RMSE about 3.5 s.



SAME MODEL, CIRCUIT AND DRIVER REMOVED

R^2 **0.02**

It knew Monaco laps are slow. It knew nothing about tyres.

DUPLICATED RACES

inflated an earlier R^2 to 0.948

ROUND_NUMBER CORRUPTED

in 82% of training rows

TYRE AGE BACKWARDS

-0.0045 s per lap older

GIVE EVERY DRIVER *THEIR OWN PACE.*

Each driver in each race gets a base pace, so tyres and fuel are learned only from changes inside that race. That was the step v1 was missing.

01	02	03	04	05	06
BASE PACE	COMPOUND	FUEL	TYRE WEAR	TEMPERATURE	TRAFFIC
per driver, per race	soft / medium / hard	per kg on board	by tyre age	track temp × age	car ahead within 1 s
free	3 levels	0.0333 s	6 knots	faster wear	+0.41 s
car + driver + track today	global → circuit → race	per kg per 90 s lap	0, 4, 10, 18, 28, 40 laps	when the track is hotter	per lap; +0.20 s at 1-2 s

```
lap time = base[driver, race] + L · ( compound + 0.0333·fuel + wear(age) + age·(temp + circuit + race) + traffic )
```

One penalised least-squares fit, about 2,500 parameters, solved in closed form. Gaussian priors shrink thin data toward the circuit, then the global value. L scales everything to a 90 s lap. Lap-to-lap noise: 0.41 s.

— THE TRICK

THE PIT STOP SPLITS **FUEL FROM WEAR.**

Inside one stint, fuel falls and tyre age rises together, lap for lap. You cannot tell them apart.

At a pit stop the tyres reset to zero but the fuel keeps falling. That break is where the model learns each effect.

0.0333 s/kg

About 0.058 s per lap of fuel burned: inside the range quoted for current F1 cars.



WHAT THE TYRES DO

WEAR, MEASURED.

Time lost per lap against a fresh set of the same compound, fuel removed. Fitted on all seasons.



SOFT

+1.72 s

per lap at 30 laps old

MEDIUM

+2.00 s

per lap at 30 laps old

HARD

+1.59 s

per lap at 30 laps old

TRAFFIC

+0.41 s

per lap, within 1 s of a car

— WHAT A STOP COSTS

22.6 SECONDS. *UNLESS THE SAFETY CAR IS OUT.*

GREEN FLAG



22.6 s

100% of a green stop

VIRTUAL SAFETY CAR



13.3 s

59% of a green stop

SAFETY CAR

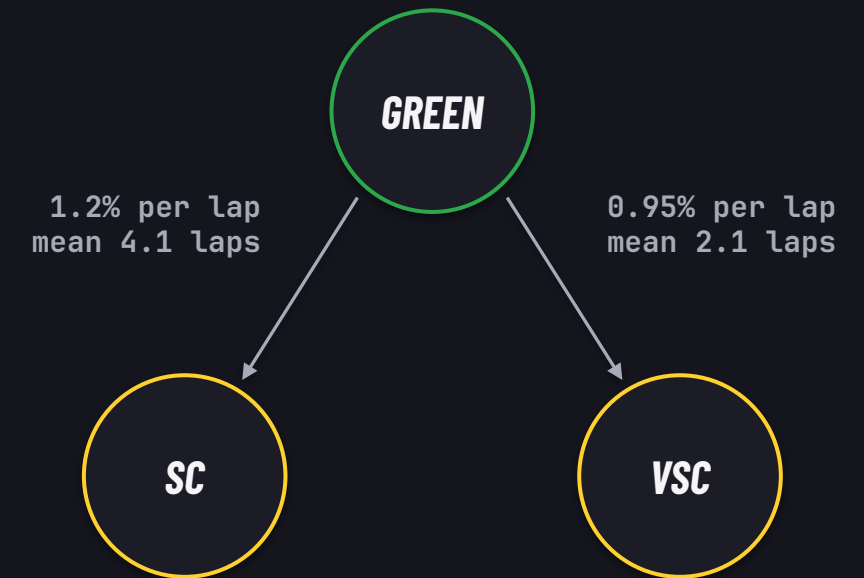


1.9 s

9% of a green stop

↑ how teams treat it: about 50%

SAFETY CAR RISK · PER-CIRCUIT MARKOV CHAIN

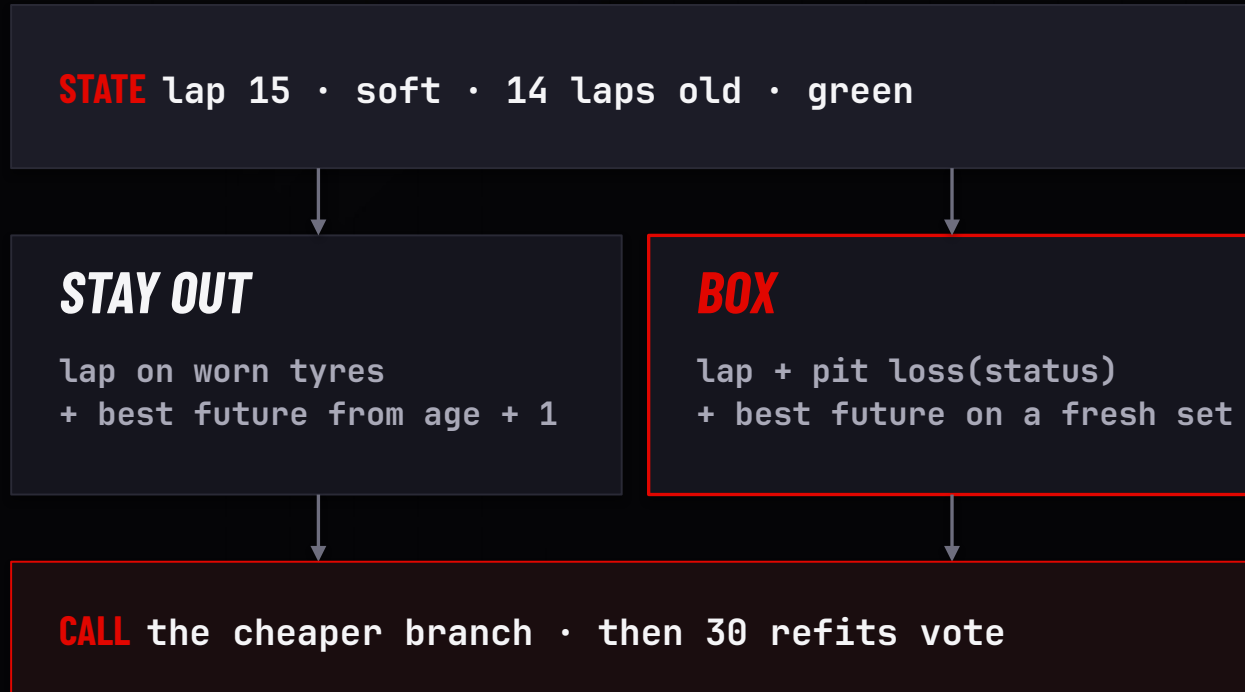


Shrunk toward the global rate with a 200-lap prior.

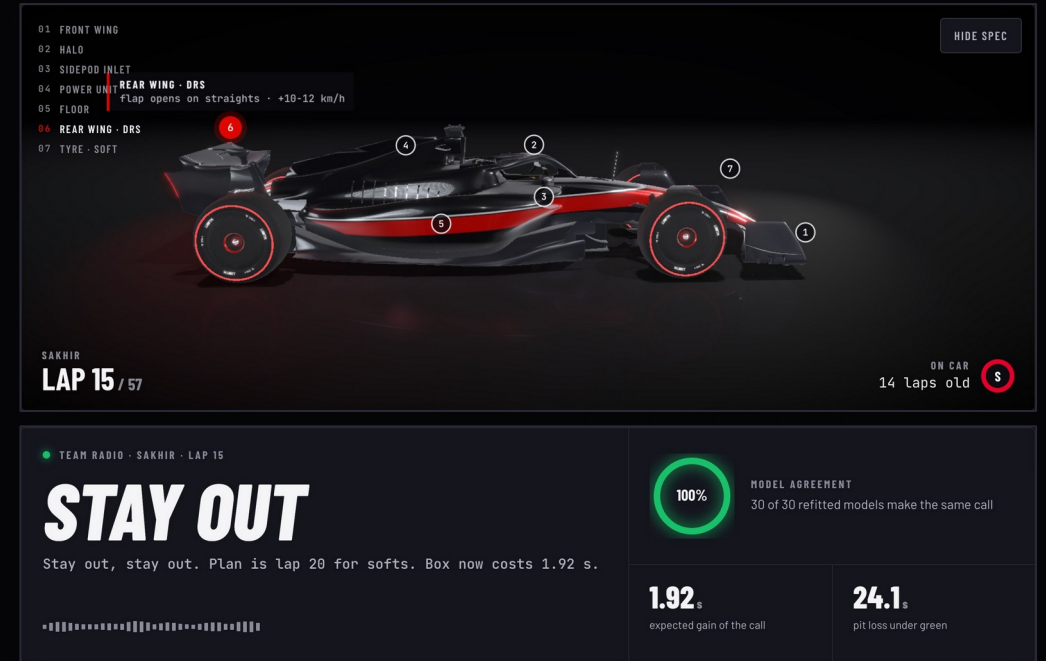
Green: in-lap + out-lap minus the two laps the model expected. 1.53 s error per stop vs 2.16 s for a fixed 22 s (465 stops).

Safety Car: gap to the leader three laps later, against cars that did not stop.

SOLVE THE REST OF THE RACE **BACKWARDS.**



Safety Car states follow the circuit's chain, so staying out for a cheap stop has value. No finish without two compounds. Past the 95th-percentile stint, a cliff penalty instead of extrapolation.

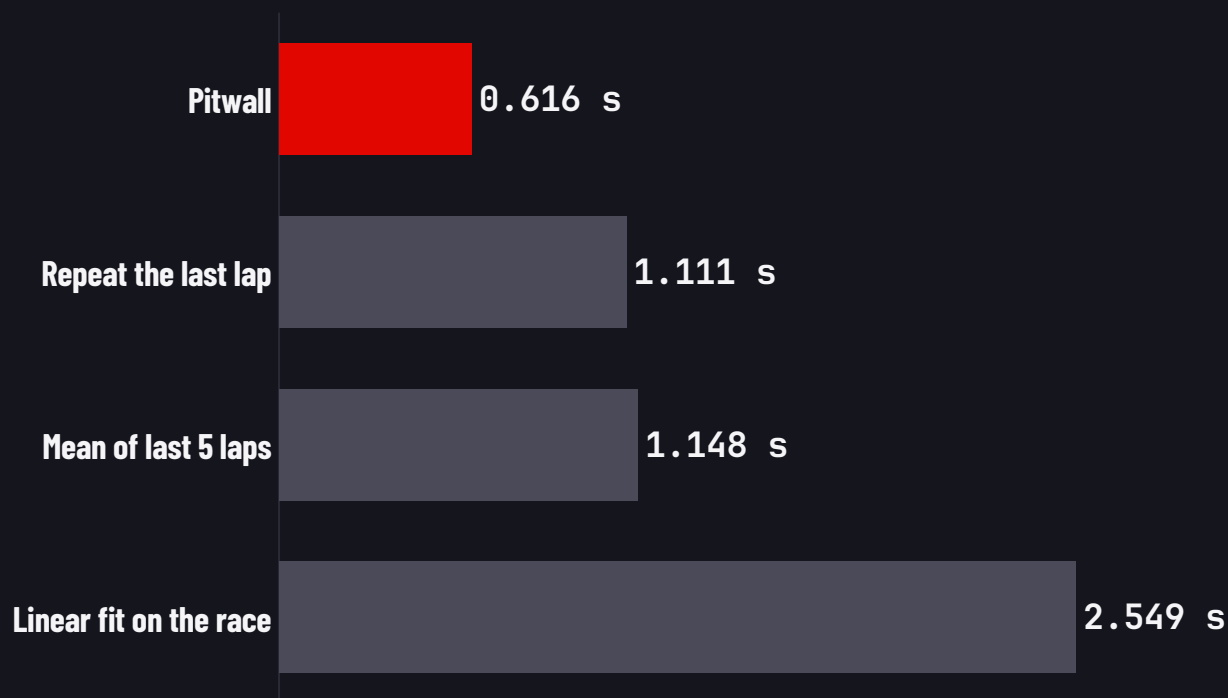


The Pit Wall view: the call as team radio, 30-model agreement, the time it is worth. The browser runs a line-for-line port of the optimiser, tested against Python.

— RESULTS · 2025, NEVER SEEN IN FITTING

45% LESS ERROR *THAN THE OBVIOUS GUESS.*

LAP-TIME ERROR DURING A RACE · SECONDS (MAE)



109,883 predictions across the 2025 season

0.446 s

1-5 LAPS AHEAD

error forecasting the next
few laps

1.556 s

BEFORE THE RACE

v1 Ridge: 1.708 s · R^2 0.955

1.53 s

PIT LOSS PER STOP

fixed 22 s: 2.16 s · 465 stops

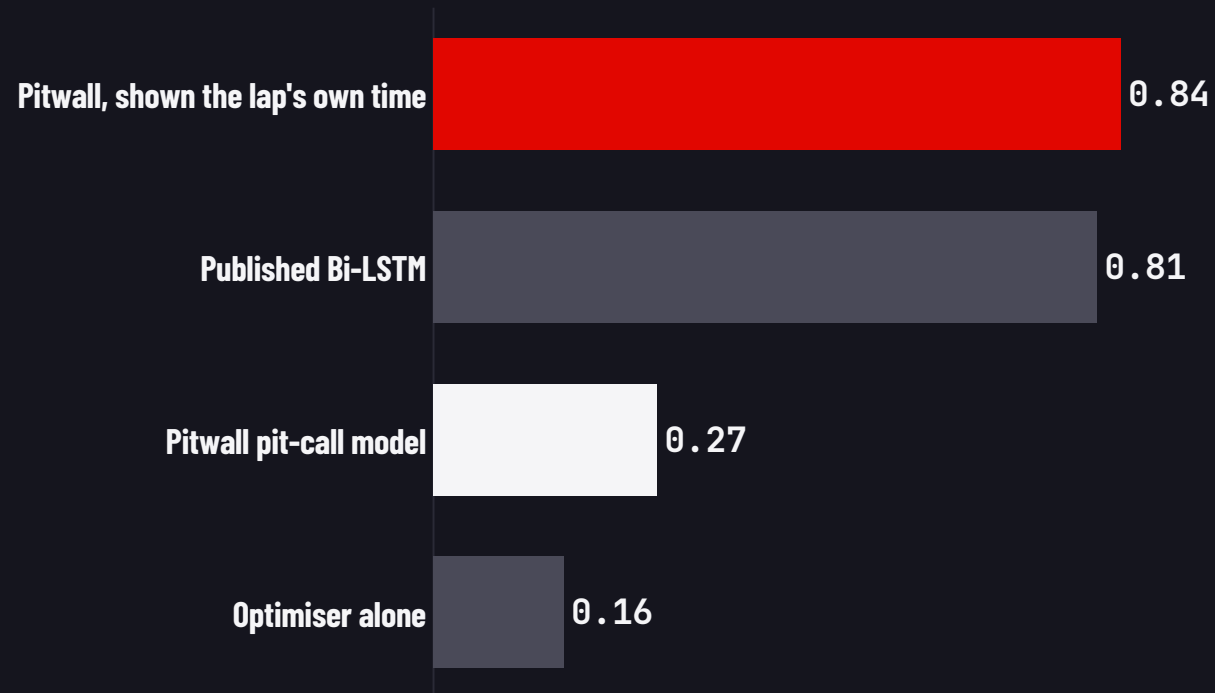
0.49%

REST-OF-RACE TIME

from lap 10 · naive: 1.32%

THE BEST-LOOKING SCORE **WAS A LEAK.**

PREDICTING REAL PIT STOPS, PER LAP · F1 SCORE



2025 season · 22,331 lap states · 625 real stops

**A lap that ends in the pit lane is slow
*because of the stop.***

Give a model that lap's time and it reads the answer off the lap.

LEAKY

F1 0.84 · ROC-AUC 0.994

HONEST

F1 0.27 · ROC-AUC 0.876

Our own leak: end-of-lap gap features lifted F1 from 0.27 to 0.32 until they were moved to lap n-1.

DID NOT *FINISH.*

#	SYMPTOM	CAUSE	FIX	
1	<i>R² 0.948 THAT WOULD NOT REPRODUCE</i>	popular races counted several times in the data	deduplicated; honest baseline R ² 0.91	FIXED
2	<i>RACES MIXED TOGETHER</i>	round_number corrupted in 82% of training rows	race IDs rebuilt from file names	FIXED
3	<i>NO PIT LAPS TO LEARN FROM</i>	FastF1 IsAccurate drops in-laps, out-laps and SC laps	keep every lap; flag representative ones	FIXED
4	<i>SILENT GARBAGE IN THE FIT</i>	groupby returned index labels, not positions	factorise + argsort split	FIXED
5	<i>SOLVER CRASHED MID-FIT</i>	SciPy positive-definite solve segfaulted	numpy.linalg.solve	FIXED
6	<i>PIT LOSS NONSENSE UNDER SC</i>	the Safety Car queue gives the time back	gap to leader vs matched non-stoppers	FIXED
7	<i>F1 0.32, TOO GOOD</i>	end-of-lap gap leaked the in-lap	features from lap n-1: honest 0.27	FIXED

AND ON *THE TRACK.*

#	SYMPTOM	CAUSE	FIX	
1	3D VIEW CRASHED WITH NAN	service worker served a stale tracks.json	network-first for data; cache-first only for hashed files	FIXED
2	RACE REPLAY FROZE FOR GOOD	one exception inside requestAnimationFrame	schedule the next frame first	FIXED
3	CHASE CAM 100S OF METRES BEHIND	position lerp at 64× playback	rig derived from the car's own pose	FIXED
4	CAR FLIPPED IN CORNERS	rotation degenerate near a 180° heading	roll-free yaw + pitch poses	FIXED
5	CAMERA WOBBLE AT 1×	telemetry jitter, arc-length error	smoothed centreline, damped shared heading	FIXED
6	FRONT-WING LABEL NEVER DREW	one of seven Html roots failed to mount	one projected DOM overlay	FIXED
7	CHICANES MISSING	smoothing flattened the geometry	corners from braking zones in the speed trace	FIXED

— HOW IT EVOLVED

FROM NOTEBOOK *TO PIT WALL.*

2026-07-10

V1



FastF1 pull, dedup and splits; Ridge baseline on lap time

2026-09-03

V1.X



per-compound models; tyre effect barely visible; round_number fix

2026-10-06

V2



structural model, measured pit loss, SC chain, DP optimiser

2026-10-06

V2.1



recency-weighted in-race fit (−18% error); leak-free pit-call model

2026-10-06

V2.2



cinematic 3D app: real car model, race theatre

2026-10-06

V2.3



F1 broadcast redesign, What If, onboard, guide, corner cinematics

WHAT EACH STEP TAUGHT

ABLATE FIRST

NEGATIVES COUNT

MODEL THE BASE

LEAKS HIDE IN TIME

MAKE IT WATCHABLE

MAKE IT STABLE

Same data from July to October. What changed was the question asked of it: not “how fast is this lap?” but “how much slower than this driver’s own pace today, and why?”

THE PRODUCT

SIX VIEWS. ONE RACE ENGINEER.

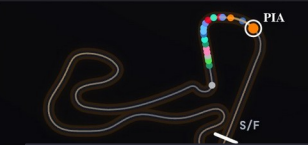
LAP 1 / 72
DUTCH GRAND PRIX

DIRECTOR CHASE TV HELI OVERVIEW

PIA · MCLAREN

C9 RIGHT · 65°
apex 256 km/h · 6th



**RACE THEATRE · DIRECTOR CAMERA**

01 FRONT WING
02 HALO
03 SIDEPOD INLET
04 POWER UNIT
05 FLOOR
06 REAR WING · DRS
07 TYRE · SOFT

HIDE SPEC



PIT WALL · SPEC HOTSPOTS

ON CAR
14 laps old

+4.0
+2.0
+0.0

Hard +2.9
Medium +2.5
Soft +2.4

TYRE LAB · LIVE CURVES

TYRE AGE (LAPS) 28°C

Pit Wall · Race Theatre · Onboard · What If · Tyre Lab · Proof

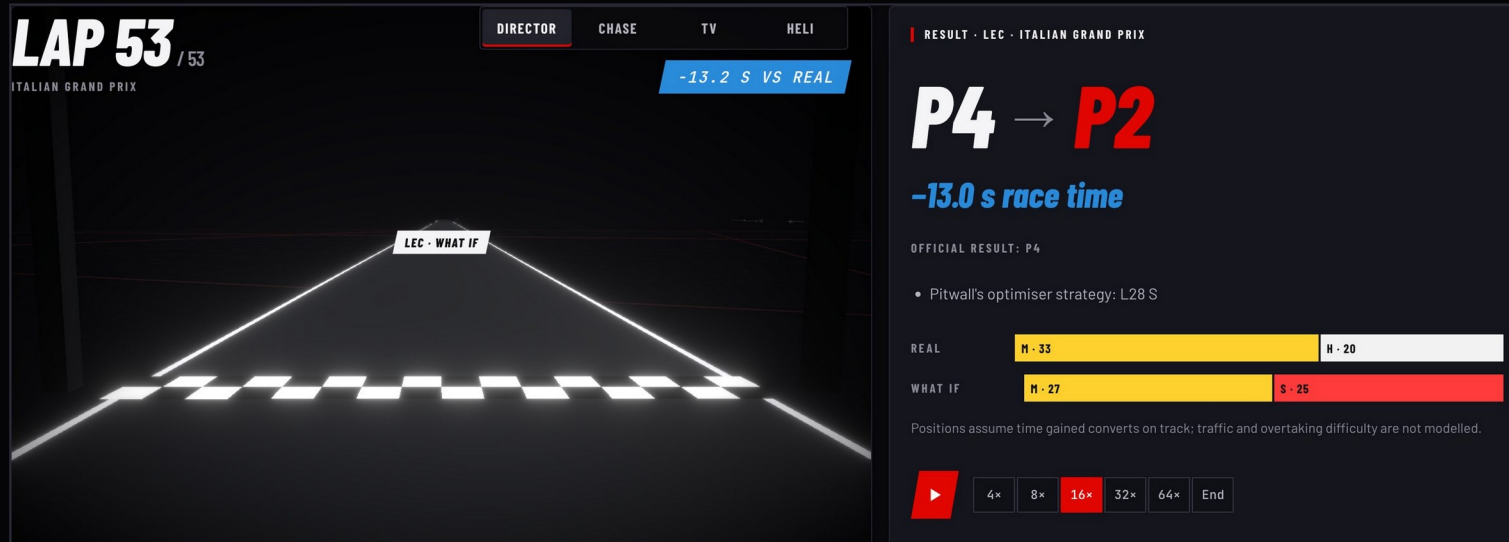
Live: b25bb1004-wq.github.io/F1-Pitstop_Strategy

— WHAT IF

CHANGE ONE THING. **RERUN THE RACE.**

What if Leclerc had followed Pitwall's strategy at Monza?

RUN IT



RESULT

P4 → P2

RACE TIME

-13.0 s

STRATEGY

lap 28 for softs

01

PARSE

Deterministic. Anything it cannot place is reported, never guessed.

02

REAL PACE

Each driver's clean pace, lap by lap, pit-lane time removed.

03

APPLY

Tyre curves, pit loss and the optimiser change the laps that differ.

04

SAFETY CARS

Queue events; lapped cars keep their deficit.

05

ANCHOR

what-if clock = real clock + (what-if - baseline).
No change = reality.

— MAKING IT FEEL REAL

DIRECTED LIKE A BROADCAST.



Corner caption, apex speed and gear come from the real fastest lap.

CORNERS FROM BRAKING

$\Delta v \geq 20 \text{ km/h}$

Monza: all 6 braking zones found

LONG-LENS TRACKSIDE CAMS

$\text{fov} = 2 \cdot \text{atan}(7.5 \text{ m} / d)$ outside each corner, zooming to hold the car

STEERING FROM CURVATURE

$\delta = \text{atan}(3.6 \text{ m} \times \kappa)$

front wheels turn into each corner

ROLL-FREE POSES

yaw + pitch only

the car can never flip

DIRECTOR CUTS

cut at every corner

chase, side and aerial on straights; steady at 8+

FIVE THINGS ***THIS TAUGHT ME.***

01 ***A HIGH R^2 CAN BE A LOOKUP TABLE.***

Ablate before you believe it: 0.91 became 0.02.

02 ***SEPARATE EFFECTS WHERE THEY DECOUPLE.***

The pit stop resets the tyres, not the fuel.

03 ***LEAKS HIDE IN TIME.***

Use only what was known before the lap.

04 ***HOLD OUT WHOLE SEASONS.***

Tune on 2024, score 2025 once.

05 ***STABILITY IS A FEATURE.***

Smooth inputs and damped rigs beat clever cameras.

WHAT'S *NEXT.*

01

TWO-CAR UNDERCUT MODEL

The optimiser minimises one car's own time. Undercuts and covering rivals are why real calls differ. The gap features already exist.

02

C1-C5 ALLOCATION

Soft, medium and hard are relative per event. Mapping them to Pirelli's compounds would separate the curves.

03

FP2 LONG RUNS

Very old tyres are mostly seen on cars where they held up. Practice long runs counter that survivorship bias.

Known limits today: What If assumes time gained becomes positions (traffic and overtaking are not modelled). Wet races are excluded. Before a race, a new season's car pace is the main unknown.



CHEQUERED FLAG

BOX, BOX.

LIVE APP

b25bb1004-wq.github.io/F1-Pitstop_Strategy

CODE

github.com/b25bb1004-wq/F1-Pitstop_Strategy

HOW THE MODEL WORKS

[notes/how_the_model_works.md](#)

Arnav Yadav · FastF1 timing data 2021-2025 · frontend built with Claude Code · car model "Ferrari F1-75" by Sketcher, CC-BY-NC-4.0, de-branded · fonts: Barlow, JetBrains Mono (OFL)